

SMTP Traffic Analysis Report

Splunk SIEM — Security Operations Analysis

SOC REPORT | CONFIDENTIAL

1. INTRODUCTION

SMTP (Simple Mail Transfer Protocol) log files provide detailed records of email communication, including sender and recipient addresses, timestamps, and server response codes. This report presents an analysis of SMTP logs ingested into Splunk SIEM, aimed at understanding email traffic patterns and detecting anomalous or potentially malicious behavior within the network.

2. EXTRACTED LOG FIELDS

The following key fields were extracted from the raw SMTP log events to support structured analysis:

Timestamp: 1332008186.970000 — precise event time
Session ID: CTjlgc2KKuLjKtxQc — unique per connection
Source IP / Port: 192.168.202.138 : 36060 — client endpoint
Destination IP / Port: 192.168.21.102 : 25 — target mail server (SMTP)
Protocol: SMTP
Domain: example.com — target domain
Status / Error Code: 503 / 5.5.1 — authentication not enabled
Flow Direction: F — forward flow flag

3. EMAIL TRAFFIC PATTERN ANALYSIS

A Splunk SPL query was used to extract sender and recipient addresses via regex, bin events by one-minute intervals, and compute email volume alongside unique sender counts. A risk score was assigned — flagging **HIGH_RISK** when a single sender generates more than 50 emails. The query returned 194 events in a single time bin, originating from exactly 2 senders (*nessus@nessus.org* and *root@hec.net*), and was scored **NORMAL** under the current threshold logic — though the burst pattern warrants deeper inspection.

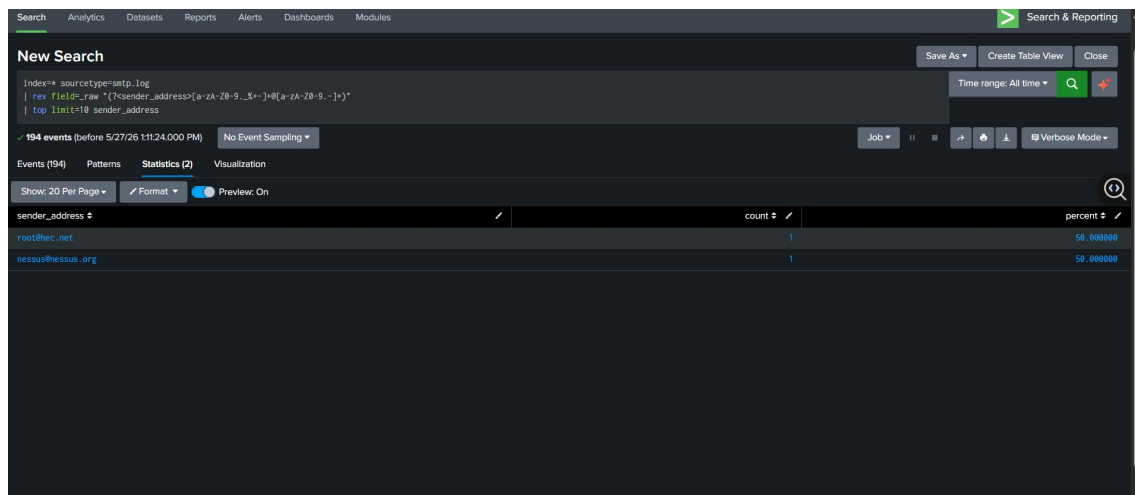


Figure 1 — Splunk query output: email volume, unique senders, sender/recipient addresses, and risk score

4. ANOMALY DETECTION

To detect unusual patterns, a *timechart* with a 1-hour span was used to plot event counts over time. All 194 events fell within a single one-hour window on the same day, confirming a concentrated burst rather than distributed activity. A separate query identified the top senders, revealing that both *root@hec.net* and *nessus@nessus.org* each accounted for exactly 50% of unique sender records — further highlighting the automated, non-human origin of the traffic.

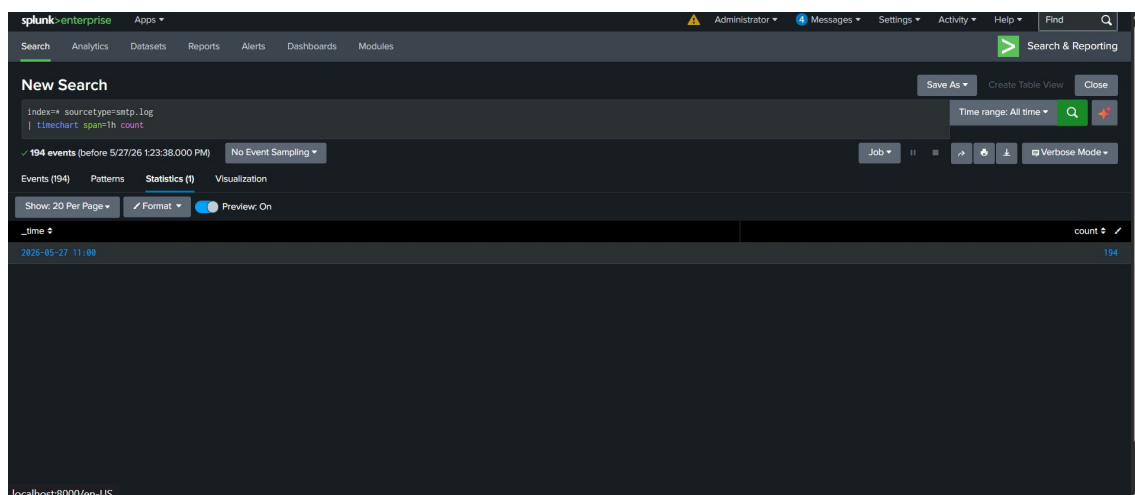


Figure 2 — Top sender distribution: root@hec.net and nessus@nessus.org, each at 50%

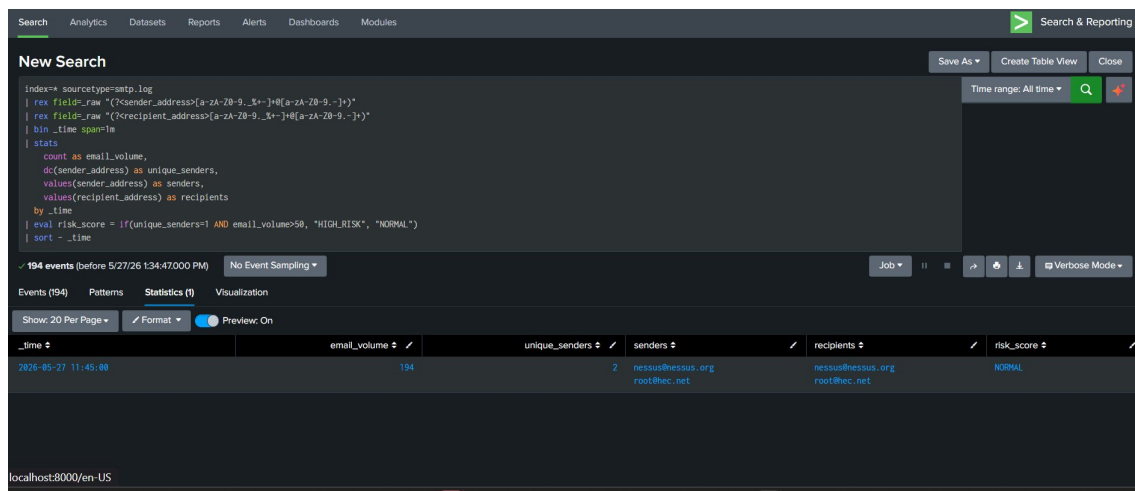


Figure 3 — Timechart (1h span): all 194 events concentrated in a single hour window

5. DEEP-DIVE FINDINGS

Closer inspection of the 194-event burst reveals the following:

Traffic Pattern: 194 emails sent in a single burst by only 2 senders — not consistent with normal enterprise email behavior.

nessus@nessus.org: Belongs to the Nessus vulnerability scanner. This tool generates automated email reports upon scan completion, producing high-volume, burst-style SMTP activity.

root@hec.net: A system or administrator-level account, likely an internal host generating automated notifications.

Risk Score Reassessment: Although the SPL eval logic flagged this as NORMAL, the automated nature of both senders means the burst is better classified as system-generated traffic requiring verification rather than clear-text normal behavior.

Two scenarios should be considered:

Scenario A (Most Likely): A scheduled Nessus vulnerability scan completed and triggered a batch of automated report/notification emails — controlled, expected, and legitimate.

Scenario B (Potential Risk): The Nessus infrastructure or the root account has been compromised or abused, with an attacker leveraging scanning tools to generate abnormal internal alerts or exfiltrate data via SMTP.

6. CONCLUSION & RECOMMENDATIONS

The observed SMTP activity is consistent with automated, system-generated traffic — most likely originating from a Nessus vulnerability scan report burst. While the volume (194 events in one hour) is anomalous in appearance, there is no strong evidence of malicious activity at this stage. The traffic pattern does not match typical spam or phishing campaigns, which would involve broader sender and recipient diversity.

The following actions are recommended:

- Confirm that a Nessus scan was scheduled and executed at the observed time.

- Verify that root@hec.net is a known and authorized internal system account.
- Establish an email volume baseline to improve future anomaly thresholds in the SPL risk score logic.
- Monitor for similar SMTP bursts originating from unknown or external senders.
- Consider tuning the HIGH_RISK threshold or adding sender-identity checks to the detection query.

"The email burst represents automated vulnerability scanning activity (Nessus), resulting in a high but controlled volume of SMTP traffic. While anomalous in appearance, it is most likely legitimate system behavior — pending confirmation of the scheduled scan."